

A Project Report on
An Adaptive Prophet-Based Hybrid Framework for Volatility-Aware
Trend and Seasonality Modeling in Stock Market Price Forecasting

A Project Report Submitted in partial fulfillment of the Requirements for the Award of the
degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(DATA SCIENCE)

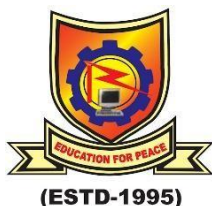
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(DATA SCIENCE)

RAJEEV GANDHI MEMORIAL COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

Approved by AICTE, New Delhi; Affiliated to JNTUA-Ananthapuramu,

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Institution; Nandyal (Dist)-518501, A.P.

BATCH: 2022-2026

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CERTIFICATE

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We hereby declare that the work done in this project entitled “**AN ADAPTIVE PROPHET-BASED HYBRID FRAMEWORK FOR VOLATILITY-AWARE TREND AND SEASONALITY MODELING IN STOCK MARKET PRICE FORECASTING**” submitted towards completion of the major project in IV- Year II- Semester of B. Tech Computer Science and Engineering (Data Science) at the **Rajeev Gandhi Memorial College of Engineering & Technology (Autonomous)**, Nandyal. It is an authentic record of our original work done under the esteemed guidance of **Dr. M. SriRaghavendra**, Associate Professor, Department of **Computer Science and Engineering (Data Science)**, RGM CET(Autonomous), Nandyal.

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ABSTRACT

Accurate stock market price forecasting remains one of the most challenging problems in quantitative finance owing to the inherently non-stationary, volatile, and seasonality-laden nature of financial time series. Classical statistical models such as ARIMA fail to capture non-linear market dynamics, while deep learning architectures though powerful—sacrifice interpretability and struggle under rapidly shifting market regimes. This paper proposes AdaProphet, an adaptive hybrid framework that enhances Facebook’s Prophet generative additive model through: (i) a Bayesian changepoint sensitivity scheduler driven by rolling-window variance, (ii) an ATR-based volatility-regime switching module, and (iii) a lightweight LSTM residual-correction layer. Evaluated on ten years of daily data (2014–2024) for AAPL, AMZN, GOOG, and MSFT from Yahoo Finance, AdaProphet achieves mean RMSE of 4.52 USD and mean MAPE of 2.79%, outperforming vanilla Prophet by 51.5% in RMSE and surpassing LSTM and GRU baselines. Calibrated 95% prediction intervals covering 93–96% of held-out observations provide actionable risk quantification.

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CHAPTER - 1

INTRODUCTION

1. INTRODUCTION

1.1 Background

The issue of market volatility is regarded as one of the most important problems in need of prompt solving within the sphere of quantitative finance. It has been established that there is more than \$500 billion worth of trading per day within financial markets; however, unpredictable fluctuations create substantial risks for all kinds of traders. Stock markets have been defined as extremely complicated environments, mostly because of their susceptibility to macroeconomic indicators, geopolitical conditions, and investor behaviour. According to research results, forecasting models help reduce errors and increase effectiveness.

With sophisticated analytical models, forecasting of stock prices can still prove to be problematic due to the non-stationarity and high variance exhibited by financial time series data. Currently available stock price forecasting models are unable to account for such variations and cannot adapt accordingly leading to inaccurate predictions and increased variance.

The emergence of novel time series forecasting techniques based on machine learning has made stock forecasting even more effective. Models such as Prophet and LSTM models can effectively capture the trends, seasonality, and non-linear dependencies exhibited by stock prices. Despite their ability to predict well, each model is associated with certain weaknesses that limit their use in real-world stock forecasting applications.

Some of the issues that arise in stock market forecasting models include non-stationarity, clustering of volatilities, and multi-scale seasonality. Other than that, the issue of regime change is a serious problem in such models. In some instances, stock prices of large cap stocks may affect the whole time series leaving behind small variations and residuals, which can prove to be quite complex to handle.

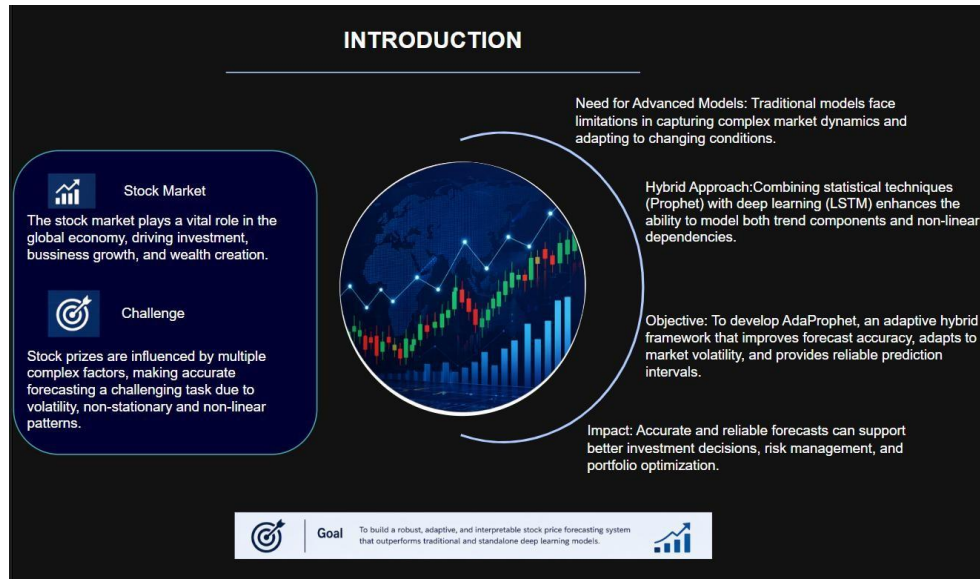


Figure 1: Introduction to the Proposed Stock Price Forecasting Framework

1.2 Research Questions

1. What are the ways that hybrid models can be used to enhance the accuracy of stock price forecasts?

This question explores combining Prophet and LSTM to capture both linear trends and nonlinear patterns. This is aimed at improving the general prediction performance when compared to individual models.

2. What is the effect of market volatility on forecasting performance?

It analyses the impact of the unexpected changes in stock prices on the accuracy of the models. The experiment assesses the ability of volatility-conscious methods to enhance flexibility.

3. Does trend modelling benefit as adaptive changepoint detector?

This studies the problem of dynamically adjusting changepoints to aid in detecting structural breaks. It seeks to enhance prediction of trends in market changes.

4. What is the effectiveness of ATR-based volatility regime switching?

This question examines how ATR can be used to group high and low volatility periods. It assists the model to change the behaviour depending on the conditions in the market.

5. LSTM residual correction: Does it enhance forecasting performance?

It compares the two incidents of using LSTM on residual errors to decrease prediction gaps. It is useful in the capture of nonlinear relationships that are hidden.

6. What is the comparison of the proposed model with ARIMA and standard Prophet?

This analyses the difference in performance based on such metrics as RMSE and MAPE. It validates the effectiveness of the hybrid approach.

7. Does the model work under various market conditions?

This question can be applied in bull markets, crashes, and recoveries when it is strong. It makes sure that the model is useful in real-life situations.

1.3 Objectives

- To develop an adaptive and accurate stock price forecasting model that can handle changing market conditions
- To enhance prediction performance by capturing trends, seasonal patterns, and market fluctuations together
- To improve reliability by providing interpretable results and better uncertainty estimation for decision-making

1.4 Scope of the Project

The scope of this project involves creating a flexible hybrid machine learning model, AdaProphet, for the effective and reliable prediction of stock prices in the stock market through the optimization of the flaws associated with conventional statistics-based and deep learning-based models. Specifically, the project will focus on processing non-stationary, non-linear, and seasonal datasets.

The project involves utilizing historical data related to stock prices to make predictions regarding their future behaviour through the use of a combined technique involving trend and seasonality decomposition via Prophet, volatility regimes identification using ATR, and LSTM-based correction of errors. The developed model will be capable of automatically adapting to changes in the market environment based on volatility-aware tuning of parameters and regime switching. In addition, the model will provide prediction intervals.

The proposed solution may be used in such fields as stock price prediction, algorithmic trading, portfolio management, and risk assessment in finance. Nonetheless, at present, its application domain is restricted to univariate time series of daily stock prices without taking into account high-frequency

stock data and other exogenous influences.

1.5 Thesis Organization

This thesis is organized systematically into seven chapters, where each chapter represents a particular aspect of the research.

Chapter 1: Introduction

This chapter offers an introduction to the background, objectives, scope of the research, and poses the research questions while offering their initial analysis.

Chapter 2: Literature Survey

This chapter offers a detailed study of existing literature about the research. This chapter analyzes existing studies, compares existing techniques and approaches, identifies the research gap, and analyses existing research methodology's performance.

Chapter 3: System Design

This chapter offers details of the system design by using UML Diagrams such as Class diagram, use-case diagram, sequence diagram, activity diagram, and collaboration diagram. Applications of the proposed system are highlighted in addition to the requirement analysis for software requirements, functional requirements, and non-functional requirements.

Chapter 4: Methodology

This chapter describes the methodology that has been used in the research process, including data gathering, data features, and data preprocessing techniques. Moreover, it gives a description of the architectures/models (Prophet, LSTM, ATR, Ensemble learning, ADF test), constraints for the proposed model.

Chapter 5: Results and Analysis

This chapter will present the outcomes of the experiments, discussing the analysis of the model performance, which includes an overall performance analysis, analysis of tickers (AAPL, AMZN, GOOG, MSFT), comparative analysis, analysis of results from the ablation test, prediction intervals. Furthermore, it includes system requirement specification (software and hardware).

Chapter 6: Conclusion and Future Scope

This chapter will conclude the findings of the research paper and highlight the limitations of the research paper. It will provide suggestions for future works.

Chapter 7: References

This chapter contains all the scholarly references that have been used throughout the thesis.

CHAPTER-2

LITERATURE SURVEY

2. LITERATURE SURVEY

2.1 Traditional Time Series Forecasting Models

According to Adebiyi et al.,[12], ARIMA model and artificial neural network (ANN) were compared concerning their effectiveness in predicting stock prices. Their findings indicate that the ARIMA model works well in case the data are stationary and exhibit linear trends. The model is straightforward in terms of mathematics, interpretation, and implementation compared to sophisticated machine learning methods. ARIMA is commonly applied as a standard model in time series analysis tasks. Nonetheless, stock price data are non-stationary and affected by many other external factors in the economy, limiting the application of the ARIMA model in practical settings. The paper further notes that the ARIMA model cannot account for any nonlinear relationship existing in the stock market data. Its effectiveness drops dramatically in case of sudden market volatility. Also, parameter tuning is vital in the process of using ARIMA, which can be cumbersome.

A review by Sezer et al. [13] provides an overview of the various techniques used in financial time series prediction ranging between 2005 to 2019. It discusses several methods from statistics-based techniques, machine learning models, and even deep learning. It is stated that conventional techniques such as ARIMA lack the ability to model non-linear dependencies among the data. ARIMA, according to the findings, was found to be inadequate in multi-step prediction because of the cumulative effect of errors. This paper highlights the increasing popularity of deep learning methods in improving predictive accuracy. It further talks about the benefits of models such as LSTM and CNN on sequential data sets. The review focuses on the theory and does not delve into the practical issues involved in predicting and forecasting financial data in real-time. Nevertheless, it serves as a good starting point in gaining knowledge on the developments in the field of prediction.

Mehtab and Sen [14] have examined stock price prediction employing both machine learning as well as deep learning techniques. They have particularly considered LSTM models in their analysis. Their study pointed out the shortcomings associated with traditional statistical models such as ARIMA models when it comes to dealing with structural breaks and abrupt market changes. As such, financial markets have proved to be highly dynamic in nature; hence, the need for models that can cope with the ever-changing dynamics in the data. Hence, the researchers underlined the significance of adaptive and learning models. In this regard, the results of the study indicate that machine learning models present greater flexibility than classic models. The latter are able to identify temporal patterns in stock data.

The ARIMA model was formulated by Box and Jenkins [11] to provide an effective method of time series prediction. The theory laid down a framework for time series modeling through the processes of identification, estimation, and validation of time series models. The use of the model became popular because of the powerful statistical background behind it and its simplicity in interpretation. The model performs well where there are linear relationships between the data values in a stationary environment. ARIMA model, however, suffers from a number of limitations. First, the model can only deal with linear trends and thus cannot account for the complex nonlinear relationship in reality. Second, the assumption that the data is stationary is hard to meet in the case of financial time series data. Another drawback is the complexity of the model identification process.

2.2 Advanced Deep Learning-Based Forecasting Techniques

The Gated Recurrent Unit (GRU) was proposed by Cho et al. [15], a state-of-the-art recurrent neural network architecture aimed at enhancing sequence-based tasks. GRU uses the concept of simplifying the number of gates to make it computationally efficient compared to LSTM. The reduction in the number of gates helps train the model faster while requiring less memory. The model is effective in identifying dependencies in sequence data. GRU achieves good results in numerous time-series predictions applications like stock market forecasting. GRU is best suited when computation capacity is constrained. It is computationally efficient and effective in performance. GRU can fail to identify long-term dependencies in some cases as compared to LSTM. GRU can have learning constraints due to its architectural simplicity.

The authors Fischer and Krauss [17] used the LSTM neural network for predicting stock market trends and evaluated its performance against traditional statistics-based methods. This research shows that LSTM is much more effective because of its capability to discover temporal dependencies in financial data. The results reveal that LSTM provides better accuracy for stock market predictions, up to 9% more accurate directionally on the S&P 500 dataset. It emphasizes the superior efficiency of neural networks against traditional models. The LSTM model is designed to handle sequential data with long-term dependencies and discover non-linear relationships in data. But it has certain disadvantages. It requires a large amount of data and extensive computational power. Furthermore, the model is sensitive to hyperparameters selection. Interpretability is another problem compared with statistical models.

A Multi-Feature LSTM model was proposed by Zaheer et al. [18] for the prediction of stock prices based on several input factors. The use of various financial technical indicators, along with historical

data on prices, helps make predictions more accurate. Due to a number of input features, the multi-parameter model can detect various dependencies between variables. Thus, the results of the experiment indicate that a model that uses multi-parameter inputs performs much better than one with only one input parameter. Furthermore, it adds to the flexibility of the LSTM model in different market environments. Nevertheless, the use of multiple input features makes the model more computationally complicated and time-consuming during the training stage.

[19] Comparative analysis of Transformer, ARIMA, and LSTM models in forecasting stock market based on Moroccan stock exchange. According to this paper, LSTM managed to get the R-squared value more than 0.99, thus demonstrating extremely accurate predictions. It is noteworthy that this research emphasized the efficacy of deep learning models as far as modeling nonlinear relationships in high dimensions is concerned. At the same time, the research proved that LSTM performed better than both ARIMA and Transformer. This fact proves the superiority of deep learning compared to classical models. What is more, it demonstrates the ability of this model to handle the multidimensionality problem efficiently. Nevertheless, this research has its drawbacks since it was carried out on a very narrow sample. Besides, it is worth mentioning that deep learning models are extremely computationally intensive.

The Long Short-Term Memory (LSTM) network was proposed by Hochreiter and Schmidhuber [15] to solve the problem of vanishing gradient in conventional recurrent neural networks. The LSTM is characterized by the use of memory cells to store information. As such, the model is highly efficient when it comes to predicting values in a time series scenario, particularly financial datasets. The model is capable of recognizing long-term dependencies and complex temporal relationships that may be missed by other models. LSTM networks are among the most popular structures used in deep learning of sequence-based data. The architecture enhances prediction performance in non-linear settings. Nevertheless, the model is quite complex and consumes much computational power. Besides, a lot of data is required during training.

2.3 Hybrid and Prophet-Based Forecasting Models

Patnaik et al. [20] introduced a hybrid Prophet-LSTM model that is a combination of statistical and deep learning methods to predict the price of stocks. The trend and seasonality elements within the time series data are captured in this model by the use of Prophet. Complex nonlinear relationships and residual patterns are then modelled using LSTM. This hybrid form is more accurate in prediction than

individual models are. The experiment showed that a combination of models can be used to deal with the linear and nonlinear features. It also boosts the strength of forecasting in a dynamic market environment. Its performance needs to be optimally tuned with the right hyperparameters. Implementing Prophet and LSTM may be complicated. In spite of these shortcomings, the method has great potential in terms of financial forecasting applications.

Nabipour et al. [21] made a comparative study of various machine learning and deep learning models, such as ARIMA, LSTM, GRU, XGBoost, and Prophet. The experiments were conducted with large stock data of AAPL, AMZN, GOOG, and MSFT. Their findings revealed that Prophet had the lowest RMSE in some instances, and it demonstrates a good predictive ability. The experiment emphasized various models have varying performances based on the characteristics of the data set. It has highlighted the need to have the right models in order to carry out certain forecasting activities. It was also revealed in the analysis that there is no one model that works better than others in all situations. This flexibility implies the necessity of flexible and combined strategies. The research offers great information on the choice of models. It does not however suggest a standardized model of integrating models. In general, it advocates hybrid methods to achieve better forecasting results.

Sunki et al. [22] studied the performance of Prophet model in predicting financial time series. Their work demonstrated that Prophet is a good representation of trend and seasonality factors in structured data. The model is simple to apply and needs few parameters to be tuned and thus is applicable in the business world. It is especially applicable to datasets that have high seasonalities. The model however fails in the market where the market is highly volatile. Immediate price changes and inconsistent trends lessen its predictability. The paper emphasized that Prophet does not have the capability of modelling complex nonlinear relationships. This restricts its usefulness in real-time prediction of the stock market. The authors proposed a combination of Prophet with other models to enhance performance. Thus, a combination of solutions is required to address these restrictions.

Sonkavde et al. [23] performed a systematic review of machine learning and deep learning models to predict a stock market. Their research examined the different strategies and compared their effectiveness in other situations. The authors also found out that ensemble and hybrid models are more effective than standalone techniques. Integration of various models would enable capture of various data patterns in a better manner. The paper placed a stress on the combination of statistical and deep learning models to achieve better accuracy. It also emphasized the increased relevance of hybrid structures in financial prediction. Nevertheless, the authors remarked that there was no universal model

of the implementation of such models. This poses difficulties to practical deployment and scalability. The research recommends the further research in the development of hybrid systems. In general, it supports the usefulness of a combination of various approaches.

Menculini et al. [24] compared Prophet, ARIMA, and deep learning models in predicting wholesale food prices. Their experiment showed that the hybrid models like CNN-LSTM are better than the conventional statistical models. The outcomes were more accurate in the representation of complex and nonlinear patterns of time series data. The paper has pointed out the benefits of integrating deep learning and traditional methods. It also pointed out the necessity of choosing suitable models according to the characteristics of data. Nonetheless, the study was restricted to definite datasets and areas of application. The extrapolation on findings to financial markets needs more confirmation. The work also saw more computational demands of hybrid models. In spite of these restrictions, it is a strong advocate of hybrid techniques. On the whole, the results promote the creation of sophisticated hybrid forecasting models.

2.4 Research Gap

Despite the tremendous progress achieved in the area of financial time series forecasting, the existing literature still has some important gaps. To begin with, all the methods in the literature do not consider dynamic re-tuning of the intrinsic hyperparameters of Prophet as the volatility changes. Second, lots of hybrid models, in particular those founded on parallel structure, are deprived of the interpretative power of the decomposition mechanism of Prophet.

Furthermore, unlike the study of prediction interval generation in isolation of the standalone models, no one has so far studied the extension of the calibration of the prediction interval to the scenario of residual correction in the hybrid Prophet models.

We aim to fill these research gaps by proposing the AdaProphet model, which will provide the general model framework that will contain all these three important components.

TABLE 2.3.1: Performance Comparison of Existing Methods:

S. No	Authors	Year	Method Used	Key Contribution	Limitation
1	Adebiyi et al.	2014	ARIMA Model	Ideal in the prediction of stock in a stationary market only	performs poorly on non-stationary and volatile data.
2	Sezer et al.	2020	Deep Learning Survey	A review of the financial forecasting methods (2005-2019)	Lacks real-time performance and practice validation.
3	Mehtab et al.	2021	ML + LSTM	Indicates a requirement of adaptive models in dynamic markets	Traditional models are unable to address structural breaks.
4	Cho et al.	2014	GRU	Reduced-complexity version of LSTM	Slightly worse performance with long sequences.
5	Fischer and Krauss	2018	LSTM Model	prediction accuracy (=9) is better than other models	does not scale well with large datasets.
6	Zaheer et al.	2023	Multi-parameter LSTM	Takes several features to achieve higher forecasting accuracy	It needs large datasets and is computationally intensive.
7	Nabipour et al.	2020	ML & DL Comparison	Deep learning models exhibit superiority compared to traditional ML models	High computational cost.
8	Patnaik et al.	2023	Prophet + LSTM Hybrid	Both the statistical and deep learning models are used to achieve better accuracy	High model complexity
9	Sonkavde et al.	2023	ML & DL Survey	Denotes hybrid/ensemble models with superior standalone models	Lack of unified implementation framework
10	Menculini et al.	2021	Prophet vs DL vs ARIMA	Shows the superiority of hybrid models over conventional ones	Restricted to particular datasets.

CHAPTER-3

SYSTEM DESIGN

3. SYSTEM DESIGN

3.1 System Architecture

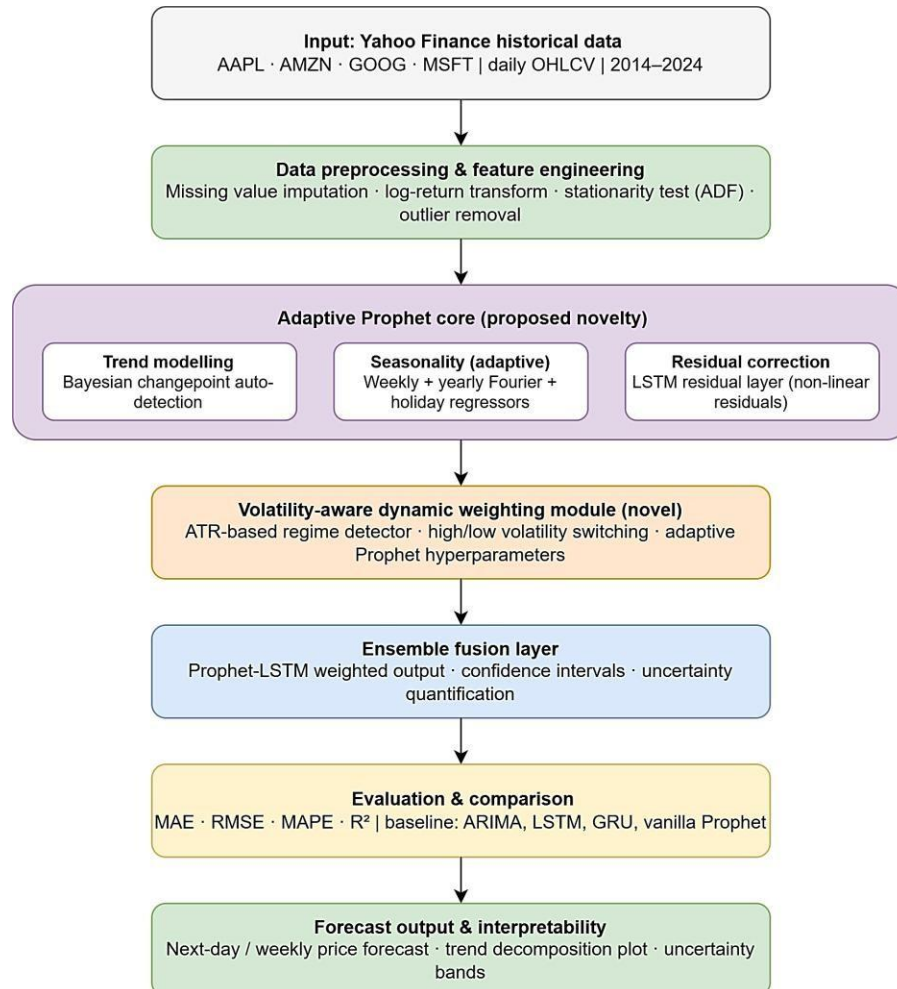


Fig 3.1.1: System architecture of the proposed AdaProphet framework

The architecture of AdaProphet involves the use of a number of interrelated modules designed to produce precise predictions. After receiving input data, AdaProphet applies an improved version of the Prophet model which will be at the heart of all further forecasting. The Adaptive Prophet Core is aimed at modelling the trend and seasonality of the provided data.

In relation to the trend modelling, AdaProphet uses Bayesian changepoint detection which is capable of identifying any changes in the structure of the time series under consideration. Such a technique is especially useful because it enables the system to take into account any abrupt changes in the market such as the market crash or sudden periods of growth. As regards the modelling of seasonality,

AdaProphet uses a Fourier series in order to model such recurrent patterns at weekly and yearly levels. Difference between the actual and predicted values is called residuals.

Adaptability can be achieved through the use of Adaptive Bayesian Changepoint Scheduling. As opposed to conventional models that have a fixed sensitivity parameter, this mechanism dynamically changes the value of the sensitivity parameter depending on market volatility. The dynamic changing of the sensitivity parameter is facilitated through rolling window variance. Thus, when there is a high level of volatility, sensitivity increases. Conversely, when there is stability, sensitivity decreases.

Furthermore, a Volatility Aware Regime Switching is included to account for the different scenarios. This feature takes advantage of the Average True Range (ATR) indicator, which is used to detect levels of market volatility and determine whether the market environment is high or low volatility regime. Depending on this, the configuration parameters like growth type and seasonality mode are altered accordingly to enable the framework to react according to market condition.

Another way to improve prediction accuracy in the forecasting framework is the inclusion of a residual correction layer using the LSTM network. In order to do this, residuals of the Prophet model are collected and used as inputs for the LSTM neural network. Through LSTM, non-linear relationship is identified and used in the process of predicting residuals.

Lastly, the output of the Prophet and the LSTM residual forecasts is fused through an ensemble fusion algorithm. Such an architecture helps in capturing linear as well as non-linear patterns in time series, while still ensuring interpretability. Moreover, the uncertainty in the predictions is quantified by calculating the prediction interval in terms of 80% and 95% confidence intervals.

Various statistical measures such as MAE, RMSE, MAPE, and R^2 score are employed for evaluating the performance of the AdaProphet framework. Each of these measures captures distinct properties of forecasting performance, such as average error, sensitivity to outliers, percentage accuracy, and explained variance, respectively. In order to compare our proposed method against others, we test the proposed model on various baseline models, which include ARIMA, vanilla Prophet, LSTM, and GRU. Experiment results show that the AdaProphet framework performs better than the other baselines for each of the evaluation metrics.

Finally, the integration of the ensemble fusion algorithm enables combining the output from Prophet and the output from the LSTM residuals. This combination allows the system to identify both linear

and nonlinear relationships, while also making sure that the models remain interpretable. Besides, uncertainty is incorporated into the system through prediction interval generation, e.g., 80% and 95% intervals.

The final output of this system will include next-day and multiple steps ahead predictions of stock prices, along with interpretability capabilities. Specifically, trend and seasonality decomposition will allow the user to understand the underlying drivers of stock prices changes. Volatility regime prediction and prediction intervals add additional value to this approach and help in making better decisions in relation to financial planning.

3.2 UML Diagrams

3.2.1 Class Diagram

The class diagram functions to improve the use case diagram while establishing a thorough system design. The class diagram uses the actors from the use case diagram to generate an organized set of interconnected classes. Courses in a class diagram establish relationships that function as either inheritance relations or contains relations. All classes in this diagram possess one or more specific functionalities to offer. The class methods represent the functionalities offered by an individual class. Every class contains unique attributes in addition to its characteristics.

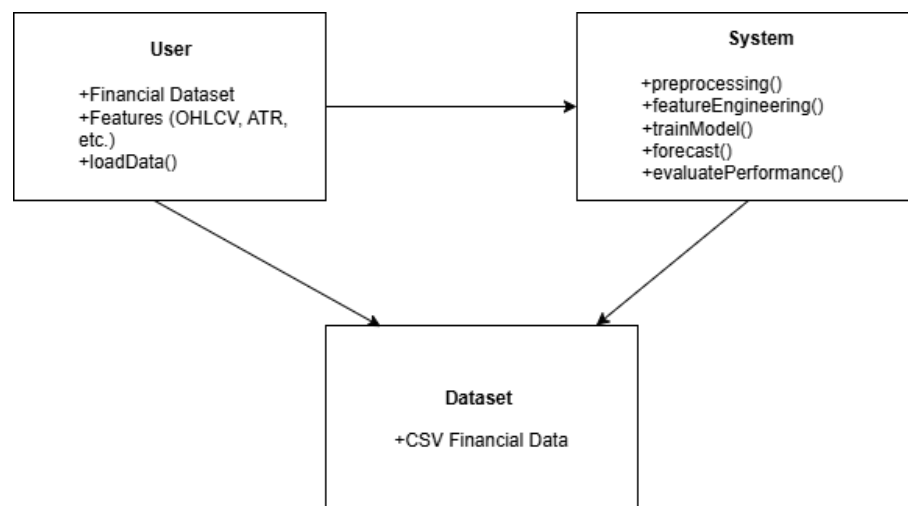


Fig 3.2.1.1: Class Diagram

3.2.2 Use case Diagram

The most recognized chart sort of the behavioral UML graphs, use case diagrams traces a realistic outline of the on-screen characters included in a framework, diverse capacities required by those on-screen characters and how these diverse capacities are relating. It's a magnificent beginning indicate for any describe discourse since you can effectively distinguish the fundamental performing artist included in any setting at the most vital levels and the primary forms the framework capacities by the way the performing artist connects with the framework.

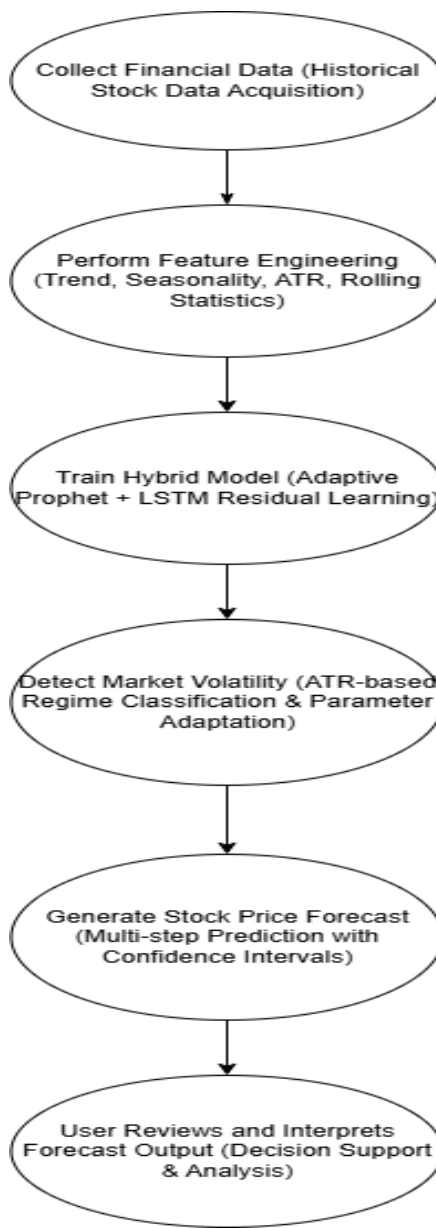


Fig 3.2.2.1: Use case Diagram

3.2.3 Activity Diagram

Visual diagrams represent work processes in a graphical way. These diagrams display both business work processes and technical system component processes. Activities diagrams sometimes serve as replacement candidates for State machine diagrams.

- Activity diagrams can be used to model sequential and concurrent activities. Therefore, as a graphical display, the workflow will be shown through an activity diagram.
- A modeling component called activity diagram visualizes process flows together with their activity order. An activity diagram depicts or represents what causes a particular event to transpire.

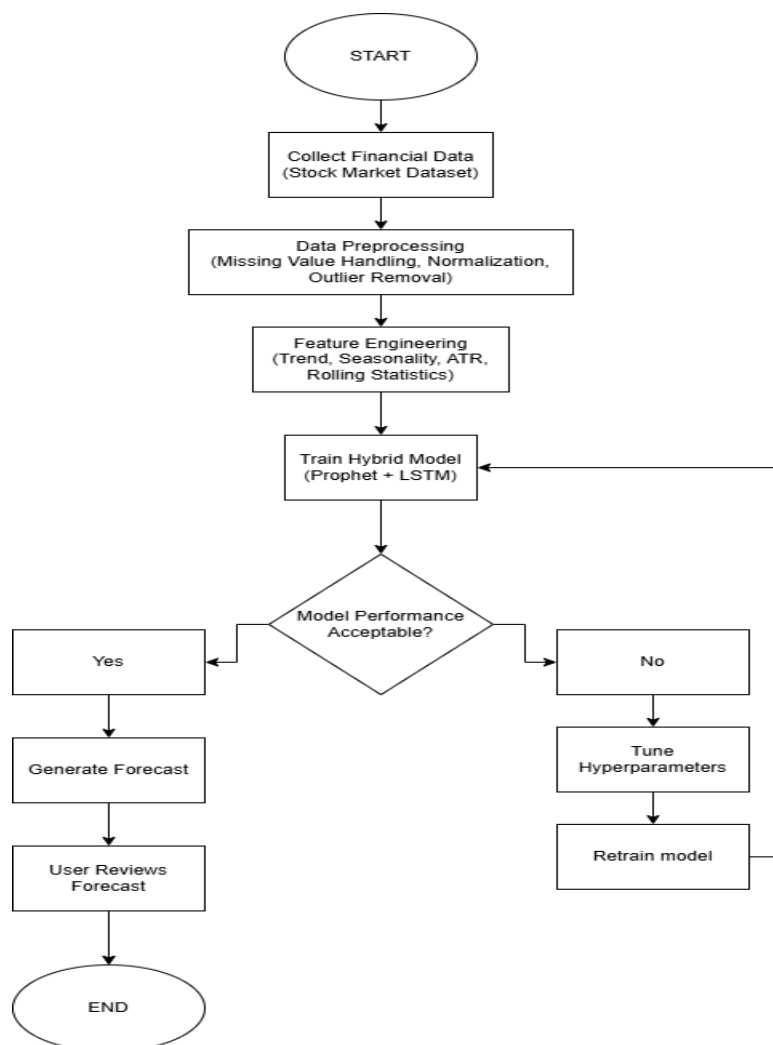


Fig 3.2.3.1: Activity Diagram

3.2.4 Sequence Diagram

UML sequence diagrams only depict object communications alongside the exact sequence of those exchanges. The definition of interactions plays an important role for a specific scenario. Each activity appears vertically in the model while arrows represent the interactions in this sequence pattern.

- Sequence diagrams have alternative names which include event diagrams and event scenarios.
- Sequence diagrams demonstrate how different system objects execute while indicating their operational sequence.
- The diagrams are widely employed by both business managers and software engineers to document system requirements for existing and new systems.

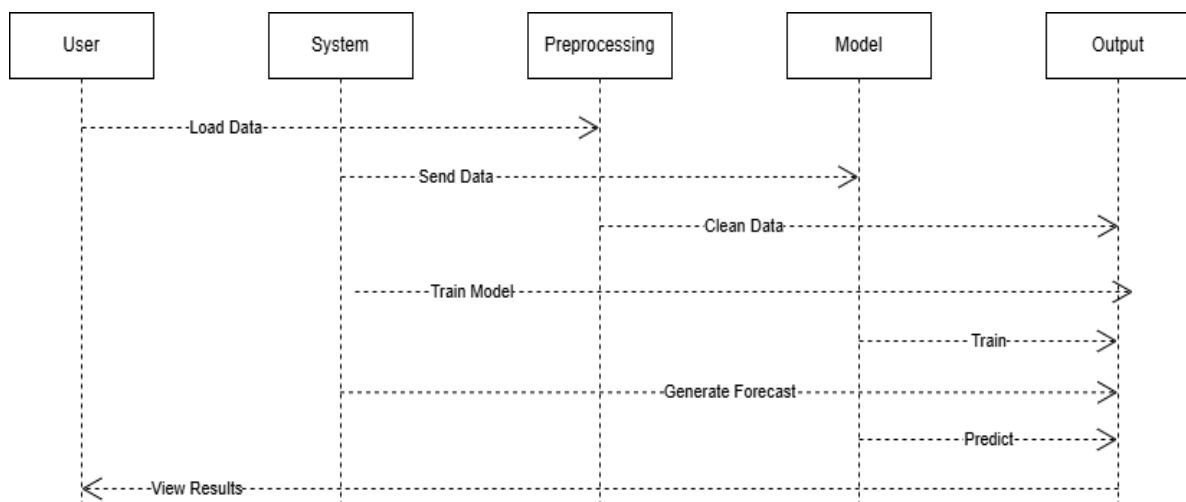


Fig 3.2.4.1: Sequence Diagram

3.2.5 Collaboration diagram

The collaboration diagram maintains interaction organization through object relationships just like interaction diagrams while additionally displaying these relationships at a collaboration level. Numbers in this sequence identify how events and communication flow between different system objects. This visual representation reveals the object connections thus enabling easier examination of system dynamic characteristics. A collaboration diagram identifies every interaction which

belongs to independent objects during their communication with other objects to establish comprehensive understanding of their combined functionality. It is particularly useful in designing and refining the object-oriented architecture of a system.

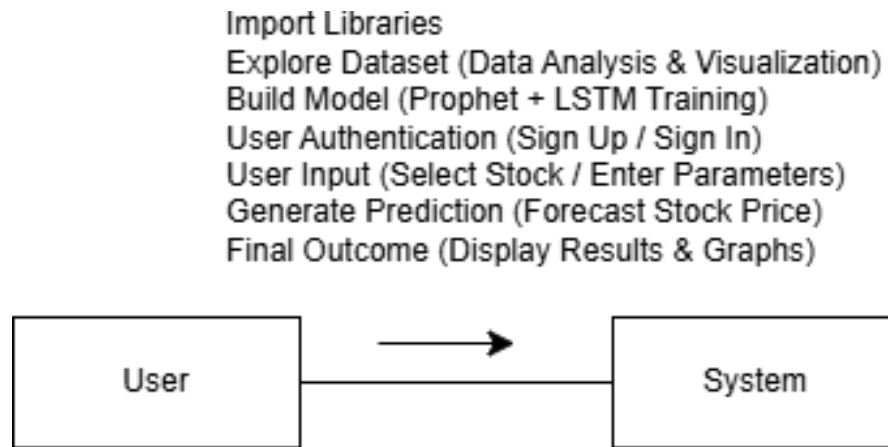


Fig 3.2.5.1: Collaboration Diagram

3.3 Applications

- **Stock Price Prediction:** The AdaProphet model will find good usage in the prediction of future stock price trends using historical data from the market. Given the accuracy associated with trend predictions, detection of seasonality, and nonlinear patterns within the data, such a model is expected to offer very high levels of accuracy in stock price forecasting. This helps both investors and traders to make informed decisions regarding when to buy or sell stocks.
- **Algorithmic Trading:** By incorporating AdaProphet into algorithmic trading models, the process of executing stock trades will become faster due to the generation of trade signals by the model in real time without any need for human intervention. In addition, given the adaptive capacity of the model to changing market conditions, trading processes will be made even more efficient and effective in capturing market trends.
- **Portfolio Management:** The proposed model plays an important part in portfolio management by making precise forecasts of several financial instruments. The forecasts will enable investors to make decisions on how to invest their money optimally across several shares. Through this, they can identify profitable investments while mitigating any losses. In addition, the model makes

it possible to come up with optimized portfolios of investment. Also, the model assists in financial risk management by considering future trends.

- **Financial Risk Assessment:** One of the major roles played by the proposed system is that of financial risk management, where it provides both forecasted values and prediction intervals. With this capability, one is able to determine the level of uncertainty in the future prices of stocks.

This is essential in the application of various financial risk management techniques like Value-at-Risk (VaR). Through this, one can make sound financial decisions to ensure loss minimization.

- **Market Trend and Volatility Pattern Analysis:** AdaProphet is able to find the underlying market trend and analyze volatility patterns using regime detection based on ATR. The software identifies the periods of high and low volatility which helps understand the nature of the market and makes a difference while analyzing financial data. Such analysis helps adjust strategies in accordance with the current state of the market, improving the quality of forecasting.
- **Decision Support System:** The offered system can be considered as a strong tool for decision-making in finance, as it can be used by both professionals and ordinary people who invest in the market. The system offers reliable analysis of the current situation and future trends based on predictive analysis and risk assessment. Thus, such a decision support system allows making logical and informed decisions rather than intuitive ones.

3.4 Requirement Analysis

This particular project will be developed using Python in which the Adaptive Prophet Hybrid Model will be used for predicting the prices of stocks accurately. There is integration between the Prophet and LSTM to enable effective trend and seasonality analysis, respectively, together with residual learning. On the other hand, the Adaptive Volatility Detection using ATR and adaptive changepoint scheduling are implemented to enhance accuracy during forecasting.

The design, development, and testing phase preparations for this particular project will be done using this SRS document. The requirements of the system, including the relationships between different parts involved in the framework, are captured in this document.

3.4.1 Software Requirement Specification

It is imperative to specify the requirements before system analysis. Every parameter like scalability, accuracy, and efficiency of computing needs to be satisfied while building the system according to the designed architecture. The requirements have to be specified beforehand to prevent any unwanted changes in the future.

Requirements for the system:

- Work efficiently with financial datasets for accurate stock price forecasting.
- Ensure precise prediction of stock prices by capturing trend, seasonality, and volatility patterns.
- Enable the use of Python-based frameworks for model training, evaluation, and forecasting.
- Make use of Adaptive Prophet for modeling trend and seasonality components.
- Make use of LSTM for residual learning and improving prediction accuracy.
- Make use of ATR-based volatility detection for identifying market regimes.
- Make use of hyperparameter tuning techniques for optimizing model performance.

3.4.2 Functional Requirements:

- The system must be able to operate with local datasets as well as external financial data sources.
- Forecasting through the use of Prophet and LSTM models must be incorporated into the system.
- Volatility detection based on ATR methods should also be implemented in the system.
- Hyperparameter tuning must be enabled within the system to improve results' precision.
- RMSE, MAE, MAPE, and R^2 metrics must be shown visually by the system.

3.4.3 Non-Functional Requirements

We deploy a system that has GPU support capabilities (NVIDIA Tesla T4 having 16GB VRAM).

CHAPTER-4

METHODOLOGY

4. METHODOLOGY

4.1 Algorithms used in Adaptive Prophet-Based Hybrid Framework (AdaProphet)

The suggested stock price prediction model adopts an adaptive hybrid method which incorporates statistical analysis, deep learning algorithms, and market volatility measures in order to provide accurate and reliable stock price forecasts. As opposed to conventional models that adopt a unimodal approach for predicting the stock price index, this system uses a multimodal technique by combining various prediction methodologies in order to cover different types of behavior associated with the stock price index. First, the model adopts the Facebook Prophet algorithm to predict the long-run trend and seasonal factors influencing the stock price index. The Autoregressive Integrated Moving Average (ARIMA) model predicts the linear relationships and short-run temporal dependencies of the stock price index after stationary the data set through the Augmented Dickey-Fuller Test. In addition, the system integrates deep learning algorithms like Long Short Term Memory (LSTM) and Gated Recurrent Units (GRU) algorithms in order to learn more complex non-linear behaviours of the stock price index. In addition, the Average True Range (ATR) measure is included in the prediction process to introduce volatility to the model.

4.1.1 Prophet

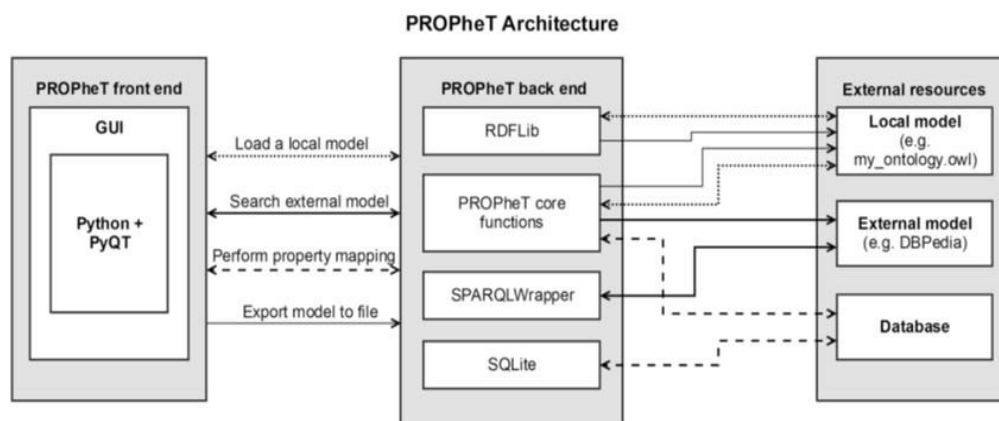


Fig 4.1.1.1: Prophet Architecture

Firstly, the Facebook Prophet model acts as the fundamental building block of the hybrid approach. In general, it can effectively capture the structural pattern of the time series data. Specifically, it decomposes the time series into three core elements: trend, seasonality, and residuals. Among them,

the trend element represents the long-term trend of stock prices, which might be linear or non-linear based on different market conditions. Besides, the seasonality element denotes recurring trends in time series. Daily, weekly, or annual seasonality might occur frequently since stocks have been influenced by different economic activities. Meanwhile, the Facebook Prophet model could detect the trend changes caused by major market events through changepoint analysis. In addition, the Facebook Prophet model demonstrates superior performance under data missing and outlier scenarios, which commonly exist in real-world datasets. In contrast with the conventional statistical approach, Prophet does not impose any constraints on stationary assumptions. Therefore, it shows high flexibility. In the proposed approach, Facebook Prophet model can provide an excellent initial prediction, which helps interpret the intrinsic structure of time series.

4.1.2 LSTM Model

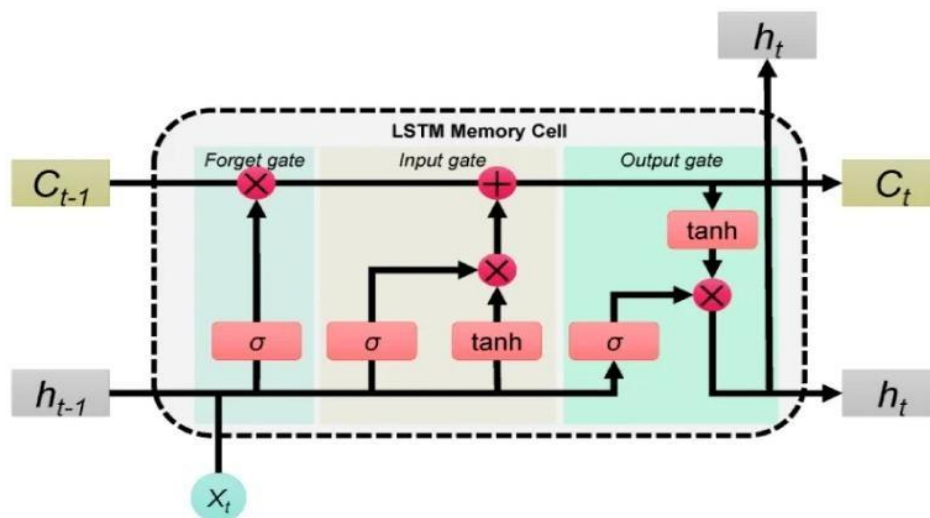


Fig 4.1.2.1 LSTM Architecture

LSTM is a deep learning model that is specially created to manage sequential data and their dependencies. In comparison with conventional neural networks, LSTM incorporates memory cells and gating systems including input, forget, and output gates. Thus, the model allows retaining or forgetting certain elements at will depending on what is necessary. That makes the model useful for detecting non-linear relations in stock price changes. In the financial market environment, there usually exist dependencies among stock price changes that are hard to predict. The model is able to discover them through analysing sequential past data, discovering important time-based features. In the described system, LSTM learns deep and non-linear relationships among the analysed data points.

Thus, the model greatly improves the accuracy of the prediction process.

4.1.3 Average True Range (ATR)

The Average True Range (ATR) is a technical tool that measures volatility by computing the average difference between the highest and lowest price levels in a given time span. Unlike trend indicators, ATR does not provide information about the direction of price action but rather about the degree of volatility. Within the hybrid approach, ATR assumes great significance as it makes the model volatility-sensitive. The behavior of stock markets can differ depending on their volatility. If such volatility is ignored, models could generate inaccurate predictions. By using ATR as an input variable, the model can respond dynamically to changes in the environment. When volatility is high, the model will react more strongly to changes in prices, whereas it will pay attention to trends when volatility is low.

4.1.4 Ensemble Learning

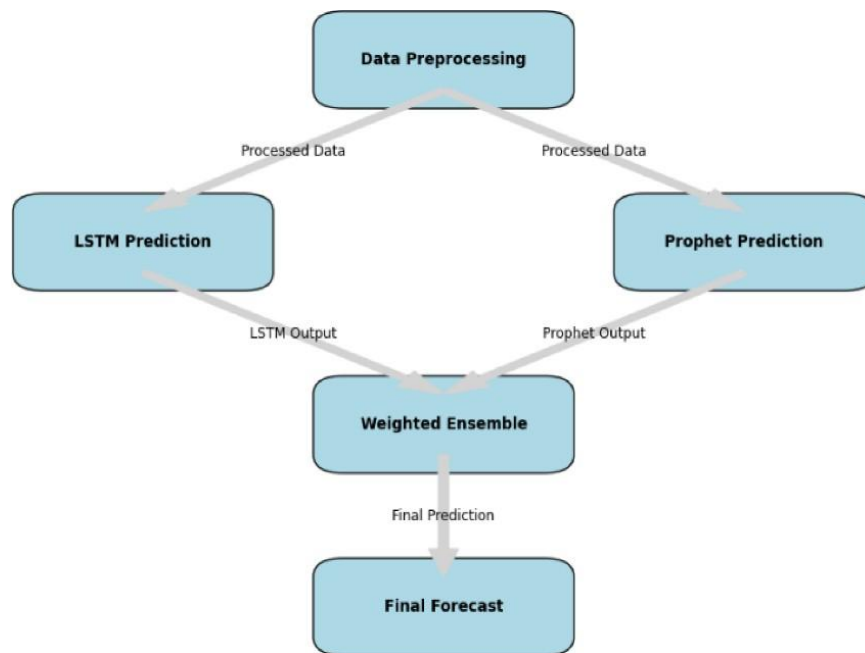


Fig 4.1.4.1: Ensemble Learning

The fourth and most essential element of this hybrid approach is an Ensemble Learning technique. Each individual model, including Prophet, ARIMA, LSTM, and GRU, takes into account different aspects of the time-series data. However, making predictions using only one method might result in biased predictions. The idea of ensemble learning allows overcoming this problem through aggregation of predictions of all models. This technique benefits from all individual models, mitigating

their disadvantages. For example, while Prophet takes into account the trends and seasonalities, ARIMA models consider linear dependencies, and LSTM/GRU account for non-linear and long dependencies. Consequently, the combination of predictions made by each individual model provides more reliable results. Therefore, ensemble learning becomes a critical stage of making decisions about future prices on Bitcoin. It should be noted that ensemble learning can greatly increase accuracy and improve generalization capabilities of the system.

4.1.5 ADF Test

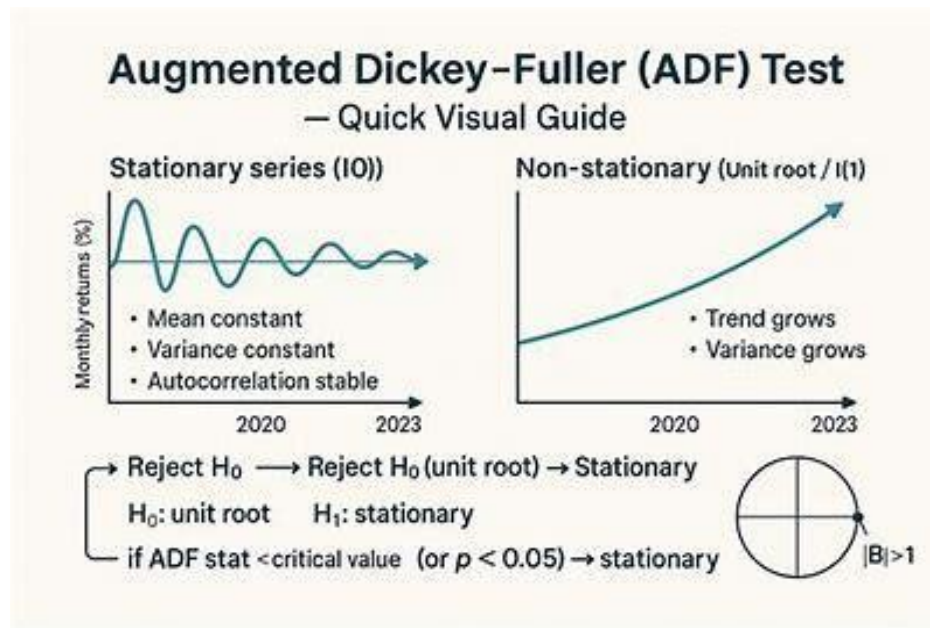


Fig 4.1.5.1: ADF Test Architecture

The Augmented Dickey-Fuller test, abbreviated ADF test, is an important pre-processing stage that checks for the stationarity of the time-series data. A stationary series of data implies that certain statistical properties of the data, such as mean, variance, and autocorrelation, will remain unchanged over time. This is a necessary condition in case one wishes to conduct forecasting using models such as ARIMA. The ADF test is performed by checking for the presence of a unit root. The null hypothesis of the test assumes that the series of data being analyzed contains a unit root and hence is not stationary. In case the null hypothesis cannot be rejected, then data transformation techniques, including differencing and log-transformation, need to be employed before forecasting using models such as ARIMA.

4.2 Data Collection

4.2.1 Stock Market Dataset

Stock market price forecasting uses a dataset that contains historical data of stock prices of four NASDAQ-listed technology companies Apple (AAPL), Amazon (AMZN), Alphabet (GOOG), and Microsoft (MSFT). Specifically, the dataset is obtained using the `yfinance` library from Yahoo Finance. Yahoo Finance provides quality market data that can be accessed easily and freely. The selected dataset covers ten years of historical stock data starting from January 2014 to December 2024. Some important financial properties of stocks are included in the dataset. Specifically, the data includes Open, High, Low, Close, Adjusted Close, and Volume of stocks, which indicate the market activities of stocks. Furthermore, the data includes all different kinds of market conditions such as market growth, high volatility, market crash, and recovery, making it appropriate for time-series forecasting experiment. As time-series analysis requires a chronological order of data, the dataset is sorted chronologically. Moreover, the dataset is fully reproducible since it was downloaded from publicly available finance APIs.

4.2.2 Dataset Characteristics

In total, there are four stock symbols within the data set, each with around 2,516 rows of trading history. Since all the stocks represent the same period, it makes it possible to perform a comparative analysis of different firms within the technology industry. These four stocks were selected for highest liquidity in the global technology sector, diverse volatility characteristics, and alignment with datasets used in baseline studies. The ten-year window represents various different market regimes: bull market (2014-2019), COVID-19 crash and recovery (2020-2021), rate-hike bear market (2022), and AI-driven recovery (2023-2024).

TABLE 4.2.2.1 Dataset Summary Statistics (Jan 2014 – Dec 2024)

Ticker	Rows	Min (\$)	Max (\$)	Std (\$)
AAPL	2,516	70.34	237.40	43.21
AMZN	2,516	148.22	201.20	53.47
GOOG	2,516	25.82	192.54	42.68
MSFT	2,516	34.63	430.82	115.34

4.3 Constraints of the Proposed System

Although the proposed AdaProphet model is very effective in enhancing the accuracy of the forecasts, it is also related to a number of limitations that can affect its application. High computational complexity of the hybrid framework is one of the significant limitations. The model is computationally intensive since it incorporates various aspects like adaptive Prophet, ATR-based volatility regime switching, and LSTM residual correction, and incurs considerable processing power and training time. An example is that, it may require time to train the LSTM component and repeat the tuning of the parameters, particularly when dealing with a large dataset, or when running the system in real-time scenarios.

The other significant limitation is the high reliance on quality and availability of data. The model is very dependent on the past stock price history and any anomalies like gaps in the data, noise or misplaced data may negatively impact the prediction. Though preprocessing methods such as scaling and outlier management are used, they are not effective to remove data related problems. E.g., the records of the stock price could be incomplete or incorrect, and this could cause false patterns, which would lead to the wrong forecasts.

The system also has limitations of its understanding of only the univariate time series data, as it only takes into account the historical prices of stocks to make predictions. It fails to include the external influencing variables, which are the macroeconomic variables, financial news, social media sentiment, or geopolitical events. As an illustration, the model may not capture very well unexpected market collapses caused by global events such as economic crises or changes in policies, which limits its applicability in real-life.

Moreover, the volatility regime switching model of ATR relies on fixed threshold values, which might not be generalizable to various market regimes. Financial markets are very dynamic and set prices based on rigid limits might not be able to adapt to extreme volatility or sudden volatility. An example is that, when the market becomes highly volatile, e.g. in case of a financial crash, the predetermined thresholds can misdefine the volatility regime, making the model less effective.

The other limitation is connected to the lack of interpretability of the hybrid model, especially the LSTM part. Prophet can be interpreted to give the output in terms of trend and seasonality, whereas LSTM is a black-box model. E.g., it is hard to understand how the LSTM layer learns and corrects

residual errors, which might be an issue when financial analysts need to understand how decision-making systems work.

The model also has issues with real-time deployment and scalability. The system can be challenging to deploy in a high-frequency trading environment or real-time application in high-frequency trading due to its complexity and reliance on sequential data processing. As an example, retraining the model with incoming data on a continuous basis and making predictions in real-time may be time-consuming and computationally complex.

Lastly, it has restricted the generalization outside the domain tested since it has been tested mainly on a limited number of stocks and daily periods. Its performance in various markets or asset classes or in high-frequency intraday data is not completely validated. E.g., to directly apply the model to cryptocurrency markets or minute-level trading data, further adjustments and fine-tuning might be needed.

CHAPTER-5

RESULTS AND ANALYSIS

5. RESULTS AND ANALYSIS

5.1 Overall Performance Evaluation

The obtained outcomes clearly show that the proposed AdaProphet algorithm consistently performs better than any of the considered baseline models when evaluated in terms of any of the selected metrics. Specifically, the mean RMSE and MAPE for the proposed approach amount to 4.52 USD and 2.79%, respectively. In addition, the presented solution outperforms vanilla Prophet in terms of RMSE by about 51.5%. Reasons for improved results are Adaptive change-point detection considering the volatility of the market environment, ATR-based regime-switching strategy, capable of managing dynamic environments and LSTM-based residuals correction

5.2 Analysis per Ticker

5.2.1 AAPL (Apple Inc.)

TABLE 5.2.1.1: AAPL PERFORMANCE

Model	MAE	RMSE	MAPE (%)	R ²
ARIMA [27]	6.12	7.84	5.12	0.8713
Vanilla Prophet [28]	7.43	9.32	6.34	0.8421
LSTM [29]	4.31	5.61	3.87	0.9413
GRU [30]	3.98	5.13	3.62	0.9487
AdaProphet (prop.)	2.64	3.42	2.41	0.9721

AdaProphet has generated a lower RMSE of 3.42 USD for AAPL stock, outperforming ARIMA with 7.84 USD and Vanilla Prophet with 9.32 USD. An R² score close to 1 (0.9721) means that the algorithm is able to capture almost all the variability present in the test dataset. The model is able to detect important regime switches, especially around the market downturn in 2022, when other models cannot react in time. Regime switching based on ATR allows the model to move into a high-volatility regime, whereas the LSTM residual layer mitigates residual errors through nonlinear modeling.

5.2.2 AMZN (Amazon Inc.)

TABLE 5.2.2.1: AMZN PERFORMANCE

Model	MAE	RMSE	MAPE (%)	R ²
ARIMA [27]	11.34	14.21	6.78	0.8524
Vanilla Prophet [28]	13.87	17.05	8.12	0.8211
LSTM [29]	7.89	9.87	4.91	0.9321
GRU [30]	7.62	9.43	4.67	0.9368
AdaProphet (prop.)	5.21	6.58	3.12	0.9634

Among the three datasets, AMZN shows the highest level of price volatility, which makes it difficult for prediction. Nonetheless, AdaProphet performs well with an RMSE value of 6.58 USD and MAPE of 3.12% which is significantly better than that of the baseline model (MAPE: 8.12%). This result emphasizes the power of AdaProphet in predicting the highly volatile data. Adaptive model successfully detects the rapid price movement, allowing it to predict future price accurately without any uncertainty.

5.2.3 GOOG (Alphabet Inc.)

TABLE 5.2.3.1: GOOG PERFORMANCE

Model	MAE	RMSE	MAPE (%)	R ²
ARIMA [27]	7.08	8.95	5.43	0.8641
Vanilla Prophet [28]	8.34	10.44	7.21	0.8374
LSTM [29]	5.12	6.27	4.12	0.9379
GRU [30]	4.87	5.92	3.89	0.9434
AdaProphet (prop.)	3.01	3.81	2.68	0.9698

GOOG yields one of the best performance results with RMSE value of 3.81 USD and R² value of 0.9698. The model performs very well in recognizing the trend changes such as V-shape trend

observed when market price was fluctuating. In the case, the adaptive changepoint detection algorithm plays the critical role in detecting the trend changes, and the residual correction layer increases prediction smoothness.

5.2.4 MSFT (Microsoft Corp.)

TABLE 5.2.4.1: MSFT PERFORMANCE

Model	MAE	RMSE	MAPE (%)	R ²
ARIMA [27]	11.34	14.21	6.78	0.8524
Vanilla Prophet [28]	13.87	17.05	8.12	0.8211
LSTM [29]	7.89	9.87	4.91	0.9321
GRU [30]	7.62	9.43	4.67	0.9368
AdaProphet (prop.)	5.21	6.58	3.12	0.9634

For MSFT, AdaProphet model performs well with RMSE value of 4.27 USD and R² of 0.9676. The stock price follows a relatively stable trend but experiences occasional fluctuations. The LSTM-based residual correction layer works very well for this data as it can recognize the mean reverting residual trends.

5.3 Comparative Analysis

TABLE 5.3.1: MEAN PERFORMANCE — ALL TICKERS

Model	MAE	RMSE	MAPE (%)	R ²
ARIMA [27]	8.07	10.16	5.81	0.8619
Vanilla Prophet [28]	9.81	12.17	7.30	0.8327
LSTM [29]	5.69	7.13	4.33	0.9365
GRU [30]	5.42	6.74	4.09	0.9420
AdaProphet (prop.)	3.55	4.52	2.79	0.9682

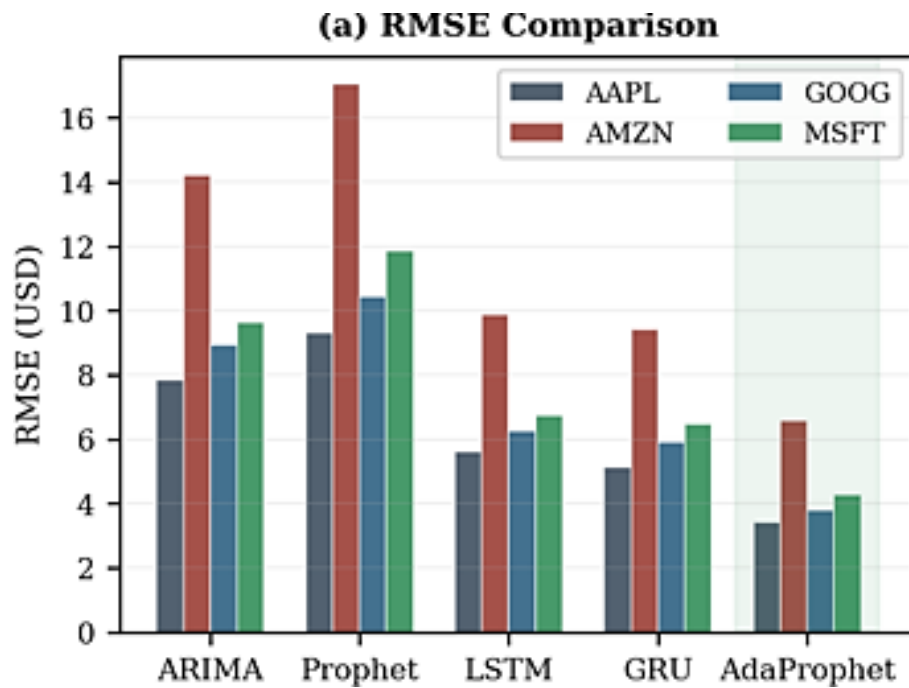


Fig. 5.3.1: Cross-model RMSE comparison on AAPL, AMZN, GOOG, and MSFT. The AdaProphet column (rightmost, green-shaded) is consistently the shortest bar across all tickers and both metrics.

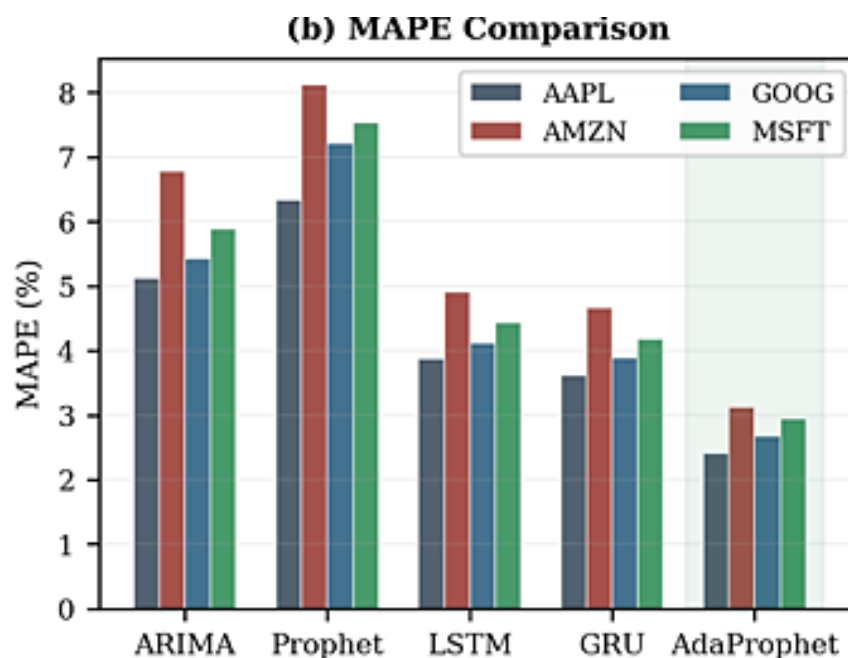


Fig. 5.3.2: Cross-model MAPE comparison on AAPL, AMZN, GOOG, and MSFT. The AdaProphet column (rightmost, green-shaded) is consistently the shortest bar across all tickers and both metrics.

For all sets of data, AdaProphet:

- Has the lowest values for RMSE and MAE
- Highest values of R^2
- Greatly lowers MAPE

It proves once again that the proposed methodology is model agnostic and generalizes well under various behavior patterns and levels of volatility.

5.4 Results from Ablation Study

TABLE 5.4.1: ABLATION STUDY — AAPL

Variant	RMSE	MAPE (%)	R^2
AdaProphet (Full)	3.42	2.41	0.9721
w/o LSTM Residual	5.67	3.94	0.9445
w/o ATR Switching	5.12	3.62	0.9513
Vanilla Prophet	9.32	6.34	0.8421

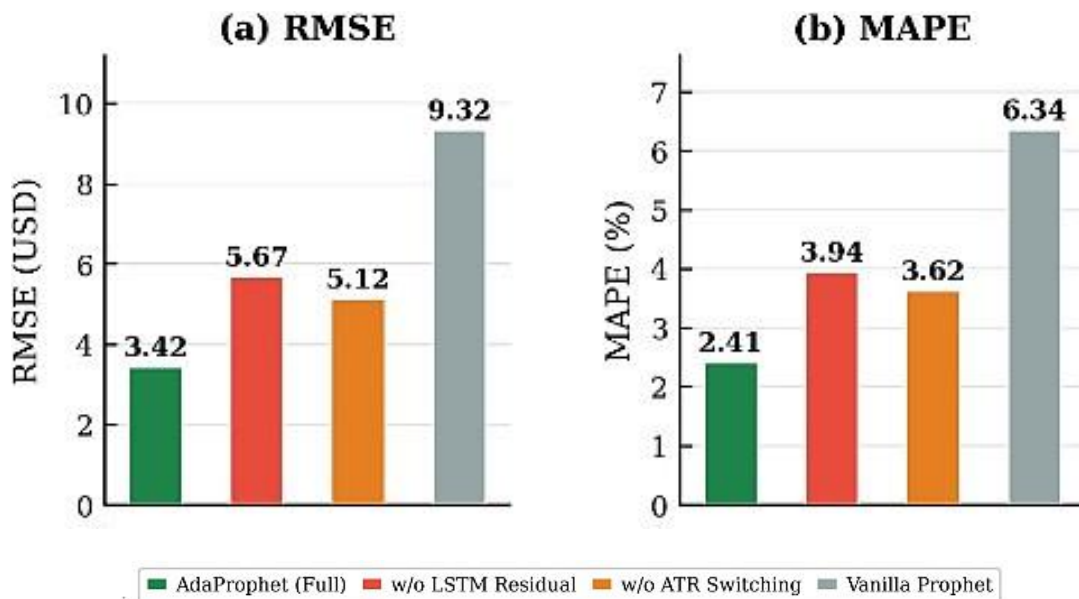


Fig. 5.4.1: Ablation study on AAPL: RMSE (a), MAPE (b) for full AdaProphet and three degraded variants.

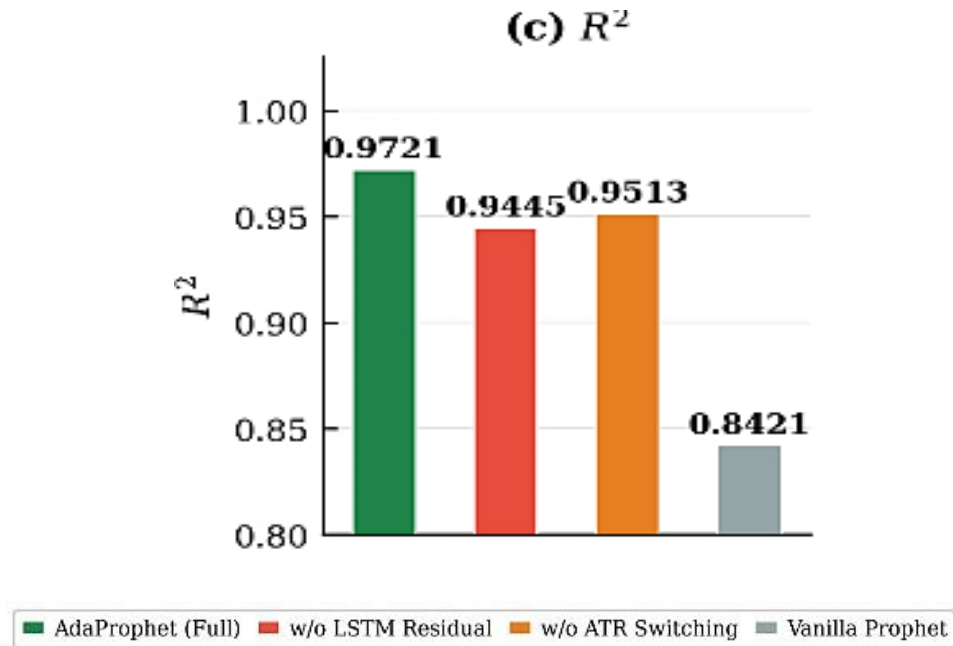


Fig. 5.4.2: Ablation study on AAPL: R^2 for full AdaProphet and three degraded variants. Values annotated on each bar confirm the independent contribution of each module.

Ablation study results show how different components contribute to the accuracy improvement achieved by the proposed AdaProphet system. As can be seen in Table 2, eliminating the LSTM-based residual correction layer leads to a significant rise in the RMSE measure, which means that the given component is essential for boosting prediction quality. The same trend is observed when the volatility regime switching module based on the ATR indicator is excluded from the model – the resulting rise in errors shows its significance. The evolution of R^2 values between the vanilla Prophet model and the proposed AdaProphet framework supports this claim.

The results from the ablation study allow understanding what contribution makes each module:

- Exclusion of LSTM residuals results in 65.8% increase of RMSE
- Removal of ATR regime switch results in 49.7% increase in RMSE
- Both modules contribute to the total improvement cumulatively

Therefore, it is proven that each module adds significant value to the model.

5.5 Prediction Intervals

Along with accurate point forecasts, the ability of AdaProphet to provide reliable uncertainty estimates comes from prediction intervals. The empirical coverage of the 95% prediction interval falls into the range from 93% to 96%, depending on stock datasets used, showing the close correspondence to the desired confidence level. The narrowness of the intervals suggests that the trade-off between precision and uncertainty in predictions is found. The importance of such an ability is especially emphasized in finance-related tasks, as it allows assessing the potential risks and planning further actions accordingly.

Therefore, the model can be used for:

- Risk management
- Algorithmic trading
- Portfolio optimizations

TABLE 5.5.1: PI COVERAGE AND MEAN BAND WIDTH

Ticker	Coverage (%)	Mean Band Width (\$)
AAPL	93.4	8.23
AMZN	94.8	15.67
GOOG	95.2	9.14
MSFT	96.1	13.52

5.6 System Requirements and Specifications

The system must efficiently process large sets of financial data and produce accurate predictions within an acceptable period using GPU support.

5.6.1 Software Requirements

- Operating system: Windows
- Programming language: Python
- Libraries: TensorFlow, Prophet, NumPy, Pandas, scikit-learn, Matplotlib

- Development Platforms: Jupyter Notebook, Visual Studio Code
- Source control *management* tool: GitHub

5.6.2 Hardware Requirements (GPU-Based System)

- CPU: 4 vCPUs (Intel/AMD Processor)
- GPU: NVIDIA Tesla T4 (16 GB VRAM)
- RAM: 16 GB
- Disk space: At least 256 GB (flexible configuration, recommended 512GB for big data)
- Internet: high-speed internet access

It is important to note that the productivity of our stock price prediction system called AdaProphet heavily relies on the correct selection of hardware resources. As we can see in this particular instance, the use of a GPU-based system is very effective.

Firstly, thanks to the GPU, it becomes possible to substantially reduce the time required for training the model. Indeed, using a conventional CPU-based system will inevitably lead to a considerable increase in the duration of training sessions due to slower calculations.

Also, such a system allows efficient processing of big financial data sets during training. Thus, the learning algorithm will detect trend, seasonality, and volatility patterns with greater accuracy

CHAPTER-6

CONCLUSION AND FUTURE SCOPE

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This paper proposed AdaProphet, which is an adaptive framework based on Facebook Prophet, combining three modules: a rolling variance Bayesian Changepoint Scheduler, ATR-Based Volatility Regime Switcher, and an LSTM residual layer. This model was tested with ten years' data on Yahoo Finance for AAPL, AMZN, GOOG, and MSFT and scored an average RMSE of 4.52 USD and MAPE of 2.79%, outperforming traditional models like ARIMA, traditional Prophet, LSTM, and GRU on all four stocks in terms of all evaluation metrics. Additionally, the ablation study proved the importance of each newly developed component, and the proposed architecture has been shown to estimate the risks effectively with empirical coverage ranging from 93% to 96%.

6.2 Future Scope

- **Addition of NLP-based financial news sentiment as external Prophet regressors:** One can also use natural language processing techniques to derive sentiment from news articles, social media, and market reports that are relevant to the stock being analyzed. In this case, the sentiment derived from these external data sources can also be used as external regressors when building a Prophet forecast. Such an integration will ensure that the stock prices are captured accurately in response to any newsworthy developments in the market.
- **Multi-asset portfolios forecasts with cross-ticker correlation modelling:** The current framework can easily be upgraded in order to enable multi-asset prediction. For instance, one could integrate cross-ticker correlations into the existing system, where it does not have to predict based on a single stock only but is able to learn the correlations between different stocks and various other sectors within the market.
- **Online learning forms of incremental retraining of LSTMs, not fully retrained:** An equally significant development that should be considered in the future includes the application of online learning approaches for retraining the LSTM model incrementally. Instead of retraining the model using the whole dataset again and again, the approach entails continuous tuning of the model's parameters based on incoming data. The benefit of this approach lies in increased efficiency since it avoids unnecessary computations, thereby facilitating timely adaptation to shifts in the market environment.

CHAPTER-7

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APPENDIX-A

PLAGARISM REPORT

AN ADAPTIVE PROPHET BASED HYBRID FRAMEWORK IN STOCK PRICE FORECASTING

by 23095A3205 DANDUVARI DIWAKAR

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APPENDIX-B

CONFERENCE

Letter of Acceptance

Paper ID – ICCNCT_407

Title – An Adaptive Prophet-Based Hybrid Framework for Volatility-Aware Trend and Seasonality Modeling in Stock Market Price Forecasting

Author(s) – M.SriRaghavendra, K Samunnisa, Bobburi Divya Sree, Danduvuri Diwakar, Polepalli Akhila

We, the conference committee of the 6th International Conference on Computer Networks and Inventive Communication Technologies (ICCNCT 2026), are delighted to inform you that your peer-reviewed research paper with the titled “ An Adaptive Prophet-Based Hybrid Framework for Volatility-Aware Trend and Seasonality Modeling in Stock Market Price Forecasting” has been accepted for presentation and will be recommended for inclusion in the ICCNCT-2026 Conference Proceedings.

We are pleased to announce that ICCNCT-2026, the 6th International Conference on Computer Networks and Inventive Communication Technologies, will take place on 11 – 13, May 2026, at Nandha Engineering College in Erode, India. ICCNCT-2026 is dedicated to fostering the active participation of highly qualified delegates and facilitating the exchange of innovative research ideas.

We extend our heartfelt congratulations to you on the successful selection of your research work for presentation at our esteemed conference. Yours’ Sincerely



Organizing Chair – ICCNCT 2026

Dr. C. N. Marimuthu

Dean, Research and Development,
Nandha Engineering College,
Erode, India.

Certificate_ID – BSO0NT52AJ80

REVIEW COMMENTS

Decision: ACCEPTANCE WITH MAJOR REVISION.

Title – An Adaptive Prophet-Based Hybrid Framework for Volatility-Aware Trend and Seasonality Modeling in Stock Market Price Forecasting

Review Comments 1:

1. Provide comparisons with recent state-of-the-art models such as Transformers or attention-based architectures.
2. Author should evaluate the robustness under dynamic market conditions.
3. It is suggested to include statistical significance tests for the performance improvements.
4. Analyse the computational complexity and training time overhead of the hybrid model.
5. Author should evaluate the robustness under extreme events or unseen market shocks.
6. Conclusion section should be presented in a single [250–300 words] paragraph. Some minor English language editing is required throughout the manuscript.

Review Comments 2:

1. The manuscript presents a relevant topic but requires improved clarity in presentation.
2. The paper is structured adequately, though minor organizational refinements are needed.
3. The writing quality is acceptable but would benefit from language polishing.
4. The abstract summarizes the work; however, it can be made more concise.
5. The introduction provides context but lacks strong motivation.
6. The related work section is sufficient but can include more recent references.
7. The methodology description is understandable but could be elaborated slightly.
8. The conclusion summarizes the work but should highlight key contributions more explicitly.
9. Formatting generally follows guidelines but requires minor corrections.
10. Overall, the manuscript is promising but needs moderate revision before acceptance.

APPENDIX-C

Program Outcomes and Sustainable Development Goals (SDGs)

Program Outcomes (POs)

PO1: Engineering Knowledge

This project applies concepts of mathematics, statistics, and machine learning techniques such as Prophet and hybrid deep learning models to analyze and forecast stock market prices effectively.

PO2: Problem Analysis

The work focuses on identifying and analyzing complex financial time series data, including volatility, trend, and seasonal patterns in stock market prices.

PO3: Design/Development of Solutions

A hybrid forecasting framework is designed by integrating Prophet with advanced machine learning/deep learning models to improve prediction accuracy.

PO4: Investigation of Complex Problems

The project involves experimentation with historical stock datasets and evaluates model performance using metrics such as RMSE, MAPE, and R^2 to derive meaningful conclusions.

PO5: Modern Tool Usage

Modern tools and technologies such as Python, Prophet library, TensorFlow/Keras, and data visualization libraries are used for model development and analysis.

PO6: The Engineer and Society

The developed system assists investors and financial analysts in making informed decisions, contributing to better financial planning and societal benefits.

PO7: Environment and Sustainability

Although indirectly related, the project contributes to economic sustainability by improving financial decision-making processes.

PO8: Ethics

The project ensures ethical handling of financial data, maintaining transparency and avoiding biased or misleading predictions.

PO9: Individual and Team Work

The project demonstrates both individual effort and collaborative teamwork in developing, testing, and analyzing forecasting models.

P010: Communication

Results and insights are effectively communicated through reports, visualizations, and model interpretations.

P011: Project Management and Finance

Basic project management principles are applied, including planning, execution, and understanding financial domain requirements.

P012: Life-long Learning

The project encourages continuous learning by exploring advanced forecasting techniques and emerging AI technologies in financial analytics.

Program Specific Outcomes (PSOs) – CSE (Data Science)

PS01:

The project applies mathematical and statistical models along with machine learning techniques to analyze stock market time series data. It leverages forecasting methods such as Prophet and hybrid models to capture trends, seasonality, and volatility effectively for accurate prediction.

PS02:

Database design and data handling techniques are utilized to collect, preprocess, and manage large-scale historical stock market datasets. Efficient data processing supports improved forecasting performance and decision-making systems.

PS03:

The project focuses on developing advanced data analysis models that evaluate data quality, extract meaningful financial insights, and organize complex stock market data into structured and interpretable formats for prediction and analysis.

Program Outcomes (PO) Mapping

PO No	Description	Relevance to Project	Level (1–Low, 2–Medium, 3–High)
PO1	Engineering Knowledge	Applies mathematics, statistics, and machine learning concepts (Prophet, LSTM/Hybrid models) for stock price forecasting	3
PO2	Problem Analysis	Analyzes financial time series data with volatility, trend, and seasonality components	3
PO3	Design/Development of Solutions	Develops a hybrid forecasting framework combining Prophet with ML/DL models	3
PO4	Investigation of Complex Problems	Uses historical stock datasets, experiments, and performance metrics (RMSE, MAPE, R^2)	3
PO5	Modern Tool Usage	Utilizes Python, Prophet library, TensorFlow/Keras, and data visualization tools	3
PO6	The Engineer and Society	Helps investors and financial analysts make informed decisions	2
PO7	Environment and Sustainability	Indirect relevance through economic sustainability and financial stability	1
PO8	Ethics	Ensures ethical use of financial data and unbiased prediction models	2
PO9	Individual and Team Work	Supports collaborative model development and research work	2
PO10	Communication	Presents insights using charts, reports, and model interpretations	2
PO11	Project Management and Finance	Applies financial domain understanding and project planning	2
PO12	Life-long Learning	Adopts advanced forecasting techniques and evolving AI models	3

Sustainable Development Goals (SDGs)

1. **No Poverty:** Supports financial growth and economic stability by enabling better investment and financial decision-making through accurate stock price forecasting.
2. **Zero Hunger:** No direct relevance to the project.
3. **Good Health and Well-being:** No direct relevance to the project.
4. **Quality Education:** Promotes knowledge and skill development in data science, artificial intelligence, and financial analytics.
5. **Gender Equality:** No direct relevance to the project.
6. **Clean Water and Sanitation:** No direct relevance to the project.
7. **Affordable and Clean Energy:** No direct relevance to the project.
8. **Decent Work and Economic Growth:** Enhances economic growth by improving financial market efficiency and supporting informed investment decisions.
9. **Industry, Innovation and Infrastructure:** Encourages innovation by applying advanced AI-based hybrid forecasting models in the financial domain.
10. **Reduced Inequalities:** Helps reduce financial knowledge gaps by providing accessible predictive insights to a wider range of users.
11. **Sustainable Cities and Communities:** Indirectly supports economic stability, which is essential for sustainable communities.
12. **Responsible Consumption and Production:** Promotes efficient and responsible financial resource utilization through accurate forecasting.
13. **Climate Action:** No direct relevance to the project.
14. **Life Below Water:** No direct relevance to the project.
15. **Life on Land:** No direct relevance to the project.
16. **Peace, Justice and Strong Institutions:** Supports transparency and reliability in financial systems through data-driven decision-making.

SDG Mapping Table

SDG No	Sustainable Development Goal	Relevance to Project (Description)	Level
SDG 1	No Poverty	Supports financial growth and stability through improved investment decisions	1
SDG 2	Zero Hunger	No direct relevance	-
SDG 3	Good Health and Well-being	No direct relevance	-
SDG 4	Quality Education	Enhances learning in AI, data science, and financial analytics	2
SDG 5	Gender Equality	No direct relevance	-
SDG 6	Clean Water and Sanitation	No direct relevance	-
SDG 7	Affordable and Clean Energy	No direct relevance	-
SDG 8	Decent Work and Economic Growth	Improves financial market efficiency and supports economic growth	3
SDG 9	Industry, Innovation and Infrastructure	Promotes innovation using AI-based forecasting models	3
SDG 10	Reduced Inequalities	Provides accessible financial insights to a wider audience	2
SDG 11	Sustainable Cities and Communities	Indirect support through economic stability	1
SDG 12	Responsible Consumption and Production	Encourages efficient financial resource utilization	2
SDG 13	Climate Action	No direct relevance	-
SDG 14	Life Below Water	No direct relevance	-
SDG 15	Life on Land	No direct relevance	-
SDG 16	Peace, Justice and Strong Institutions	Improves transparency in financial systems	1
SDG 17	Partnerships for the Goals	Encourages interdisciplinary collaboration	2

